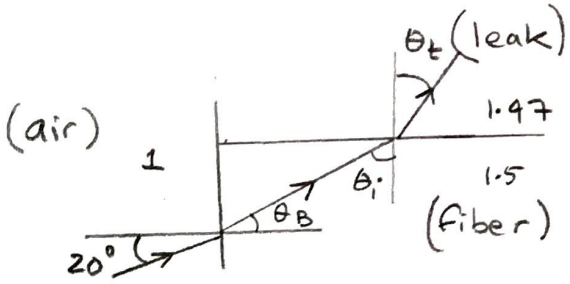
Midterm Exam Solution

① $\theta_i = 90 - \theta_B = 90 - \sin^{-1}\left(\frac{1}{1.5}\sin 20^\circ\right)$

$= 76.8^\circ$

$\theta_c = \sin^{-1}\frac{1.47}{1.5} = 78.5^\circ$



$\theta_i < \theta_c \Rightarrow$ NO TIR.

$\theta_t = \sin^{-1}\left(\frac{1.5}{1.47}\sin 76.8\right) = 83.45^\circ$

$\Gamma_{TM} = \frac{\tan(83.45 - 76.8)}{\tan(83.45 + 76.8)} = -0.324$

Leak $= T = 1 - R = 89.5\ \%$.

② $E_{photon} = 1.24/1.1 = 1.127$ eV

$P_{owar} = IA = 1\times10^{-3} \times \pi (2\times10^{-3})^2$

$\Phi = \frac{Power}{E_{photon}} = 7\times10^{10}$ photon/s

③ At 530 nm $\Rightarrow$ $n = 4.957 - j\,2.378$ ($E_{photon} = 2.34$ eV)

$v = c/n' = 0.6\times10^{8}$ m/s

attenuation $\alpha = \frac{\omega}{c}n'' = 2.82\times10^{7}$ 1/m

absorption $\alpha = 5.64\times10^{7}$ 1/m

Almost all will be absorbed when power is very low,

say 0.1 % $\Rightarrow$ $0.001 = e^{-\alpha z} \Rightarrow z \approx 122$ nm

Ge can be used, but with high reflection.

④ $n' = K/k_0 = 1\times10^{7} / \left(\frac{8\pi}{3}\right)\times10^{5} = 11.9$

$n'' = \alpha/k_0 = 0.12$

$\vec{H} = \frac{3}{\eta_0 n'}(-\vec{a}_x)\, e^{-10^5 z} \cos(8\pi\times10^{13} t - 10^{7} z)$ (A/m)

$\lambda_0 \approx 7.5$ μm (IR)